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ChE 321 HW 9
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[10-3B] $K_C = \frac{C_T}{C_I C_W}$

$C_t = C_S + C_{IS} + C_{WS} + C_{TS}$

$= C_S(1 + K_I C_I + K_W C_W + K_T C_T)$

$C_S = \frac{C_t}{1 + K_I C_I + K_W C_W + K_T C_T}$

$r_T = -r_I$

a) $W \cdot S + I \cdot S \rightleftharpoons TBA \cdot S + S$

$r_T = k_S C_{WS} C_{IS} - k_{-S} C_{TS} C_S$

$K_S = \frac{k_S}{k_{-S}} = \frac{C_{TS} C_S}{C_{WS} C_{IS}}$

$C_{WS} C_{IS} - \frac{C_{TS} C_S}{K_S} \propto \left(C_I C_W - \frac{C_T}{K_C}\right)$

$C_{IS} = K_I C_I C_S$

$C_{WS} = K_W C_W C_S$

$$\boxed{r_T = -r_I = k \frac{C_{IW} - \frac{C_T}{K_C}}{(1 + K_I C_I + K_W C_W + K_T C_T)^2}}$$

b) $I + S \rightleftharpoons I \cdot S$

$r_T = -r_I = r_{ads,I} = k_{A,I} C_I C_S - k_{-A,I} C_{IS}$

$$\boxed{r_T = -r_I = k \frac{C_I C_W - C_T/K_C}{1 + K_I C_I + K_W C_W + K_T C_T}}$$

c) $I \cdot S + W \rightarrow TBA \cdot S$

$I + S \rightleftharpoons I \cdot S,\ TBA + S \rightleftharpoons TBA \cdot S$

$I + S \rightleftharpoons I \cdot S$, $TBA + S \rightleftharpoons TBA \cdot S$

$C_t = C_S + C_{IS} + C_{TS} = C_S(1 + K_I C_I + K_T C_T \Rightarrow C_S = \frac{C_t}{1 + K_I C_I + K_T C_T}$

$r_T = -r_1 = k_S C_{IS} C_W - k_{-S} C_{TS}$

$$\boxed{r_T = -r_1 = k \frac{C_I C_W - C_T/K_C}{(1 + K_I C_I + K_T C_T)^2}}$$

d) $I + S_1 \rightleftharpoons I \cdot S_1 \Rightarrow C_{IS_1} = K_I C_I C_{S_1}$

$W + S_2 \rightleftharpoons W \cdot S_2 \Rightarrow C_{WS_2} = K_W C_W C_{S_2}$

$C_{t1} = C_{S_1} + C_{IS_1}$

$C_{t2} = C_{S_2} + C_{WS_2}$

$C_{S_1} = \frac{C_{t1}}{1 + K_I C_I}$ $\quad C_{S_2} = \frac{C_{t2}}{1 + K_W C_W}$

$I \cdot S_1 + W \cdot S_2 \rightleftharpoons TBA + S_1 + S_2$

$r_T = -r_1 = k_S C_{IS_1} C_{WS_2} - k_{-S} C_T C_{S_1} C_{S_2}$

$$\boxed{r_T = -r_1 = k \frac{C_I C_W - C_T/K_C}{(1 + K_I C_I)(1 + K_W C_W)}}$$

e) Same driving force (numerator)
With the denominator, when all 3 species absorb on the same sites, the denominator is
$(1 + K_I C_I + K_W C_W + K_T C_T)^n$
$n = 1$ or 2
When they absorb on different sites, the denominator factorizes.
Species that do not absorb don't appear in the denominator.

[10-10B)

0.1 atm $\rightarrow r \approx 0.040$

0.5 atm $\rightarrow r \approx 0.069$

1-2 atm $\rightarrow r \approx 0.060 - 0.044$

$$r \propto \frac{P_{Bu}}{(1+KP_{Bu})^2}$$

a) $$r_{MEK} = K \frac{P_{Bu}}{(1+K_{Bu}P_{Bu})^2}$$

b). gives $1+K_{Bu}P_{Bu}$ in denominator

- MEK & H_2 weakly absorbed
- squared denominator

Absorption of Bu

$$Bu + S \rightleftarrows Bu \cdot S \quad (K_{Bu})$$

Surface RXN

$$Bu \cdot S + S \xrightarrow{K} MEK + H_2 + 2S$$

Products weakly absorbed

Don't affect the site balance and can be neglected in the denominator

Let: θ_{Bu} = fraction of sites covered by Bu

θ_S = fraction of vacant sites

$$\theta_{Bu} = \frac{K_{Bu}P_{Bu}}{1+K_{Bu}P_{Bu}}, \quad \theta_S = \frac{1}{1+K_{Bu}P_{Bu}}$$

$$r_{MEK} = K' \theta_{Bu} \theta_S = K' \frac{K_{Bu}P_{Bu}}{(1+K_{Bu}P_{Bu})^2}$$

$$\boxed{r_{MEK} = K \frac{P_{Bu}}{(1+K_{Bu}P_{Bu})^2}}$$

$$\boxed{r_{MEK} = K\frac{P_{Bu}}{(1+K_{Bu}P_{Bu})^2}}$$

c) $r = K\frac{P_{Bu}}{(1+K_{Bu}P_{Bu})^2}$

$$\sqrt{r} = \sqrt{K}\frac{\sqrt{P_{Bu}}}{1+K_{Bu}P_{Bu}}$$

$$\frac{1}{\sqrt{r}} = \frac{1+K_B P_{Bu}}{\sqrt{K}\sqrt{P_{Bu}}} = \frac{1}{\sqrt{K}\sqrt{P_{Bu}}} + \frac{K_{Bu}}{\sqrt{K}}\sqrt{P_{Bu}}$$

$$\frac{\sqrt{P_{Bu}}}{\sqrt{r}} = \frac{1}{\sqrt{K}} + \frac{K_{Bu}}{\sqrt{K}}P_{Bu}$$

$$y = \frac{\sqrt{P_{Bu}}}{\sqrt{r}} \quad x = P_{Bu}$$

$$y = c + mx$$

$$c = \frac{1}{\sqrt{K}}, \quad m = \frac{K_{Bu}}{\sqrt{K}}$$

plotting y vs x should give a straight line

d) $r'_{MEK} = K\frac{P_{Bu}}{(1+K_{Bu}P_{Bu})^2}$

$F_{Bu0} = 10\ mol/min = 600\ mol/h$

$P = P_0 = 10\ atm$

stoich $1 \to 2$

$F_{Bu} = F_{Bu0}(1-X)$

$F_{MEK} = F_{Bu0}X$

$F_{H_2} = F_{Bu0}X$

$F_T = F_{Bu0}(1+X)$

$$y_{Bu} = \frac{F_{Bu}}{F_T} = \frac{1-X}{1+X}$$

$$P_{Bu} = y_{Bu}P_0 = P_0\frac{1-X}{1+X}$$

$$\frac{dF_{Bu}}{dW} = r'_{Bu} = -r'_{MEK}$$

$$X = \frac{(F_{Bu0} - F_{Bu})}{F_{Bu0}} \Rightarrow F_{Bu} = F_{Bu0}(1-X)$$

$$\frac{dX}{dW} = \frac{r'_{MEK}}{F_{Bu0}}$$

$$\boxed{\frac{dX}{dW} = \frac{k}{F_{Bu0}} \cdot \frac{P_0\frac{1-X}{1+X}}{\left[1+K_{Bu}P_0\frac{1-X}{1+X}\right]^2}}$$

$W=0,\ X=0$

to $W=23$

$X(W)$ up to 0.9

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Fitted parameters:
  k = 581.013 mol/(h·kg·atm)
  K_Bu = 2.107 1/atm
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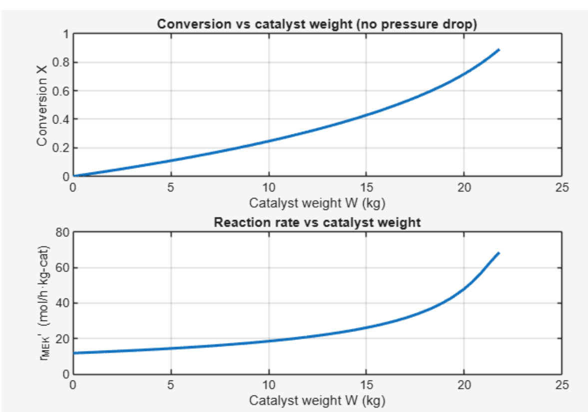

e) Suppose we want to double the MEK production without changing the temperature of the reactor or catalyst. You can only change the feed composition and total pressure. Come up with a strategy for adjusting P_0 and the inlet composition so that the overall MEK production increases while also avoiding operating conditions where the rate decreases because of Bu inhibition.

This requires critical thinking because there's no single formula that directly gives the answer.

f) $Bu \rightarrow MEK + H_2$

$$\varepsilon = \Delta n\, y_{Bu,0} = (2-1)\cdot 1 = 1$$

$$F_{Bu} = F_{Bu0}(1-X)$$

$$(1+\varepsilon X) = F_{Bu0}(1+X)$$

$$F_{Bu} = F_{Bu0}(1-X)$$

$$F_T = F_{Bu0}(1+\varepsilon X) = F_{Bu0}(1+X)$$

$$y_{Bu} = \frac{F_{Bu}}{F_T} = \frac{1-X}{1+X}$$

$$P = P_0 p$$

$$P_{Bu} = y_{Bu} P = P_0 p \frac{1-X}{1+X}$$

$$r'_{MEK} = k \frac{P_{Bu}}{(1+K_{Bu}P_{Bu})^2}$$

$$\frac{dX}{dW} = \frac{r'_{MEK}}{F_{Bu0}} = \frac{k}{F_{Bu0}} \frac{P_0 p \frac{1-X}{1+X}}{\left[1+K_{Bu}P_0 p \frac{1-X}{1+X}\right]^2}$$

$$\boxed{\frac{dp}{dW} = -\alpha \frac{1+\varepsilon X}{p}}$$

$$\varepsilon = 1$$

$$\frac{dp}{dW} = -\alpha \frac{1+X}{p}$$

$$X(0) = 0 \qquad p(0) = 1$$